Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 28

Digital Signature Schemes-II

Estimated Reading Time: 15–20 minutes

Introduction

Digital signature schemes enable blockchain networks to verify transaction authorization without requiring users to reveal their private keys. Different cryptographic algorithms provide this functionality, depending on the blockchain platform and its security requirements.

Common Digital Signature Schemes

A. RSA Digital Signature

RSA is a public-key cryptographic algorithm that can be used for digital signatures. It uses a public key for verification and a private key for signing.

RSA signatures can provide strong security when suitable key sizes, secure padding schemes and correct implementations are used. However, RSA is not the standard signature scheme for many widely used public blockchains.

B. Digital Signature Algorithm (DSA)

DSA is a public-key algorithm specifically designed for digital signatures. It allows a user to sign data with a private key and enables others to verify the signature using the public key.

DSA is historically important in digital signature technology, although newer schemes are often preferred in modern applications.

C. Elliptic Curve Digital Signature Algorithm (ECDSA)

ECDSA is based on elliptic curve cryptography and is used by several blockchain systems, including Bitcoin and Ethereum's externally owned accounts.

It provides digital signatures with relatively small keys compared with traditional RSA systems offering comparable classical security levels.

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D. Schnorr Signatures

Schnorr signatures are based on discrete logarithm cryptography and are known for their relatively simple design and useful aggregation properties in suitable protocols.

Bitcoin supports Schnorr signatures through its Taproot upgrade. Signature aggregation and related benefits depend on the protocol and implementation being used.

E. EdDSA

EdDSA is a digital signature scheme based on Edwards-curve cryptography. Ed25519 is a widely used implementation of EdDSA.

It offers efficient signing and verification and is used in various cryptographic systems, including some blockchain platforms.

Digital Signatures and Blockchain Transactions

A typical blockchain transaction authorization process involves the following steps:

1. The user prepares the transaction data.
2. The wallet constructs the required message or transaction digest according to the blockchain protocol.
3. The private key is used to generate the signature.
4. The signed transaction is broadcast to the network.
5. Nodes verify the signature and check other transaction rules.
6. A valid transaction may be included in a block through the blockchain's consensus process.

A valid signature alone does not guarantee transaction acceptance. The transaction must also satisfy requirements such as sufficient funds, correct formatting and protocol-specific validation rules.

Challenges and Security Considerations

Digital signature security depends on appropriate algorithms and careful key management.

- Private Key Theft: An attacker who obtains a private key may be able to authorize transactions.

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- Weak Randomness: Poor random-number generation can expose private keys in certain signature schemes.
- Implementation Errors: Incorrect cryptographic implementations can create vulnerabilities.
- Phishing Attacks: Users may unknowingly sign malicious transactions.
- Quantum Computing: Large-scale fault-tolerant quantum computers could threaten widely used RSA and elliptic-curve signature schemes, motivating research into post-quantum alternatives.

Applications of Digital Signatures in Blockchain

Digital signatures are used in cryptocurrency transfers, decentralized applications, smart-contract interactions, document verification, digital identity systems and blockchain-based supply-chain platforms.

Conclusion: Digital signature schemes enable secure transaction authorization in blockchain systems. Understanding algorithms such as ECDSA, Schnorr and EdDSA, together with private-key security and transaction verification, is essential for understanding blockchain security.

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